SYSTEM BLOCK DIAGRAM

ChryslerScanner V240 | integrated microSD | 259 components | 154 connected nets

VEHICLE CONNECTION

OBD connector + protection

GND
J1962-11
J1962-12
J1962-14
J1962-15
J1962-16
J1962-2
GND
J1962-11
J1962-12
J1962-14
J1962-15
J1962-16
J1962-2
J1962-3
J1962-6
J1962-7
J1962-9
N_J1_5
N_Q1B_3
J1962-3
J1962-6
J1962-7
J1962-9
N_J1_5
N_Q1B_3

File: 06_obd-protection.kicad_sch

Battery / GND

Vehicle power input

+12V
GND
+12V
GND
J1962-16
N_J1_5
J1962-16
N_J1_5

File: 07_vehicle-input.kicad_sch

Protected supply

BUS INTERFACES

CCD interface

+3.3V
+5V
GND
J1962-11
+3.3V
+5V
GND
J1962-11

File: 01_ccd.kicad_sch

Bus data

J1962-3
RX1
TBEN
TX1
J1962-3
RX1
TBEN
TX1

PCI / J1850

+12V
+3.3V
+5V
+7.5V
GND
+12V
+3.3V
+5V
+7.5V
GND

File: 02_pci.kicad_sch

J1962-2
N_Q1B_3
PCI_RX
PCI_TX
J1962-2
N_Q1B_3
PCI_RX
PCI_TX

SCI stages

SCI TX14 / TX9

+3.3V
+5V
GND
J1962-14
+3.3V
+5V
GND
J1962-14

File: 03_sci-14-9.kicad_sch

J1962-9
TX14_EN
TX2
TX9_EN
J1962-9
TX14_EN
TX2
TX9_EN

SCI TX6 / TX12 + RX bias

+3.3V
+5V
GND
J1962-12
J1962-6
+3.3V
+5V
GND
J1962-12
J1962-6

File: 04_sci-6-12.kicad_sch

RX7_BIAS
RX7_BIAS_EN
TX12_EN
TX2
TX6_EN
RX7_BIAS
RX7_BIAS_EN
TX12_EN
TX2
TX6_EN

SCI RX7 / RX15

+3.3V
+5V
GND
J1962-15
J1962-7
+3.3V
+5V
GND
J1962-15
J1962-7

File: 05_sci-rx.kicad_sch

RX15_EN
RX2
RX7_BIAS
RX7_EN
RX15_EN
RX2
RX7_BIAS
RX7_EN

Enables / bias

CONTROL & SENSING

ESP32 + RF

+3.3V
ACT_LED
EN
GND
IO0
LOG_BUTTON
MCU_SD_CS
MCU_SD_MOSI
MCU_SD_SCK
N_R25_1
PCI_RX
PCI_TX
PWR_LED
RX0
+3.3V
ACT_LED
EN
GND
IO0
LOG_BUTTON
MCU_SD_CS
MCU_SD_MOSI
MCU_SD_SCK
N_R25_1
PCI_RX
PCI_TX
PWR_LED
RX0

File: 16_mcu.kicad_sch

UART / reset

RX1
RX2
RX_LED
SCL
SDA
SD_DETECT
SD_MISO
TX1
TX2
TX_LED
V_BA
V_BT
V_PR
RX1
RX2
RX_LED
SCL
SDA
SD_DETECT
SD_MISO
TX1
TX2
TX_LED
V_BA
V_BT
V_PR

SPI / detect

I2C

I/O expander + I2C

+3.3V
+5V
GND
RX15_EN
RX7_BIAS_EN
RX7_EN
SCL
SDA
TBEN
TX12_EN
TX14_EN
+3.3V
+5V
GND
RX15_EN
RX7_BIAS_EN
RX7_EN
SCL
SDA
TBEN
TX12_EN
TX14_EN

File: 18_control.kicad_sch

TX6_EN
TX9_EN
VBB_TO_TX12
VBB_TO_TX14
VBB_TO_TX6
VBB_TO_TX9
VPP_TO_TX12
VPP_TO_TX14
VPP_TO_TX6
VPP_TO_TX9
TX6_EN
TX9_EN
VBB_TO_TX12
VBB_TO_TX14
VBB_TO_TX6
VBB_TO_TX9
VPP_TO_TX12
VPP_TO_TX14
VPP_TO_TX6
VPP_TO_TX9

Status

AW9523B | bus enables and programming control

ADC

Voltage monitoring

GND
J1962-16
VBB
VPP
GND
J1962-16
VBB
VPP

File: 17_adc.kicad_sch

V_BA
V_BT
V_PR
V_BA
V_BT
V_PR

HOST & LOGGING

USB + reset / boot

+3.3V
+5V
EN
GND
+3.3V
+5V
EN
GND

File: 19_usb.kicad_sch

IO0
N_R25_1
RX0
IO0
N_R25_1
RX0

J2 USB-C | CP2102 bridge | reset / boot

Integrated SD logger

+3.3V
+5V
GND
LOG_BUTTON
MCU_SD_CS
+3.3V
+5V
GND
LOG_BUTTON
MCU_SD_CS

File: 22_sd-logger.kicad_sch

MCU_SD_MOSI
MCU_SD_SCK
SD_DETECT
SD_MISO
MCU_SD_MOSI
MCU_SD_SCK
SD_DETECT
SD_MISO

J4 microSD | SD_3V3 | LOG / STOP

Status LED drivers

+3.3V
+5V
ACT_LED
GND
+3.3V
+5V
ACT_LED
GND

File: 20_leds.kicad_sch

PWR_LED
RX_LED
TX_LED
PWR_LED
RX_LED
TX_LED

POWER GENERATION & SCI PROGRAMMING

5 V / 3.3 V supplies

+12V
+3.3V
+12V
+3.3V

File: 08_power-5v.kicad_sch

+5V
GND
+5V
GND

VPP boost supply

+12V
+5V
+12V
+5V

File: 09_power-vpp.kicad_sch

GND
VPP
GND
VPP

VBB supply

GND
VBB
GND
VBB

File: 10_power-vbb.kicad_sch

VPP
VPP

7.5 V supply

+12V
+7.5V
+12V
+7.5V

File: 11_power-7v5.kicad_sch

GND
GND

SCI TX6 VBB / VPP drivers

GND
J1962-6
VBB
GND
J1962-6
VBB

File: 12_program-6.kicad_sch

VBB_TO_TX6
VPP
VPP_TO_TX6
VBB_TO_TX6
VPP
VPP_TO_TX6

SCI TX9 VBB / VPP drivers

GND
J1962-9
VBB
GND
J1962-9
VBB

File: 13_program-9.kicad_sch

VBB_TO_TX9
VPP
VPP_TO_TX9
VBB_TO_TX9
VPP
VPP_TO_TX9

SCI TX12 VBB / VPP drivers

GND
J1962-12
VBB
GND
J1962-12
VBB

File: 14_program-12.kicad_sch

VBB_TO_TX12
VPP
VPP_TO_TX12
VBB_TO_TX12
VPP
VPP_TO_TX12

SCI TX14 VBB / VPP drivers

GND
J1962-14
VBB
GND
J1962-14
VBB

File: 15_program-14.kicad_sch

VBB_TO_TX14
VPP
VPP_TO_TX14
VBB_TO_TX14
VPP
VPP_TO_TX14

Separate VBB / VPP rails feed the SCI programming stages; enables come from AW9523B.

Solid arrows: data and user interface paths. Dashed arrows: power or control relationships.

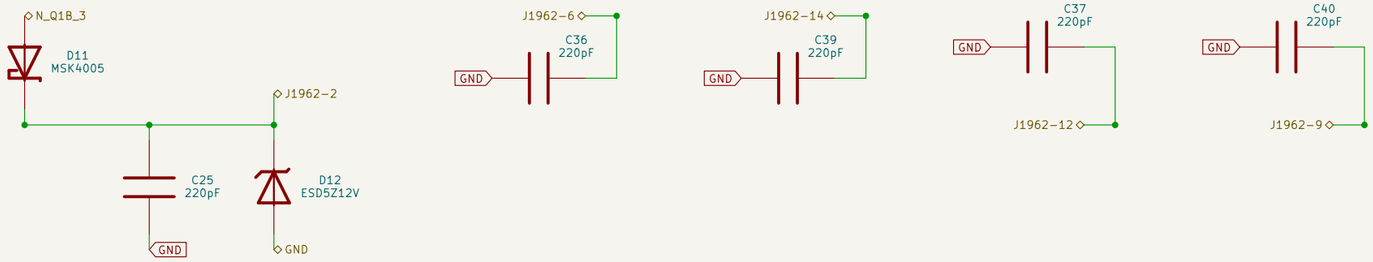
Double-click a block to open its circuit. Native hierarchical pins and named stubs carry the exact design nets.

J1 and input protection share one sheet. Solid arrows show data flow; native named stubs define electrical connectivity.

ChryslerScanner / atopile	
Sheet: /	
File: ChryslerScanner_V240_SD.kicad_sch	
Title: ChryslerScanner V240 — system block diagram	
Size: A1 | Date: 2026-10-21	Rev: V240-SD
KiCad E.D.A. 10.0.5	Id: 1/22

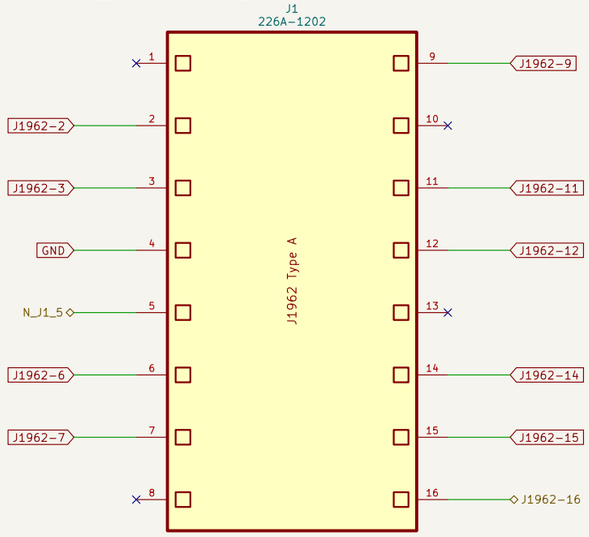
OBD connector and line protection

Matching signal names connect between sheets. Values and package pins follow the atopile netlist.





J1 — VEHICLE INPUT CONNECTOR



ChryslerScanner / atopile

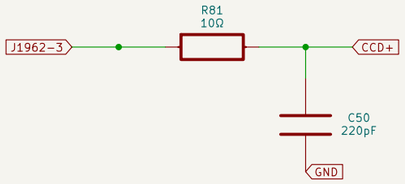
Sheet: /OBD connector + protection/
File: 06_obd-protection.kicad_sch

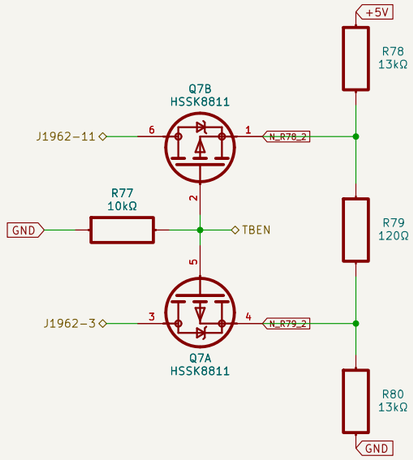
Title: OBD connector and line protection

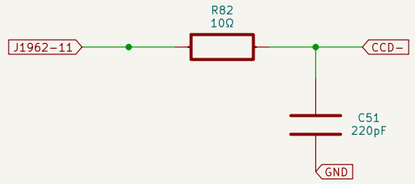
Size: A3	Date: 2026-10-13	**Rev: V240-C/OBD**
KiCad E.D.A. 10.0.5		Id: 2/22

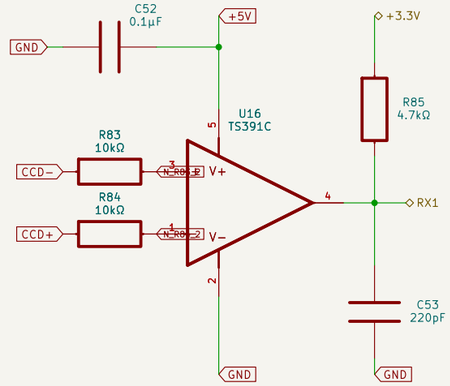
CCD interface

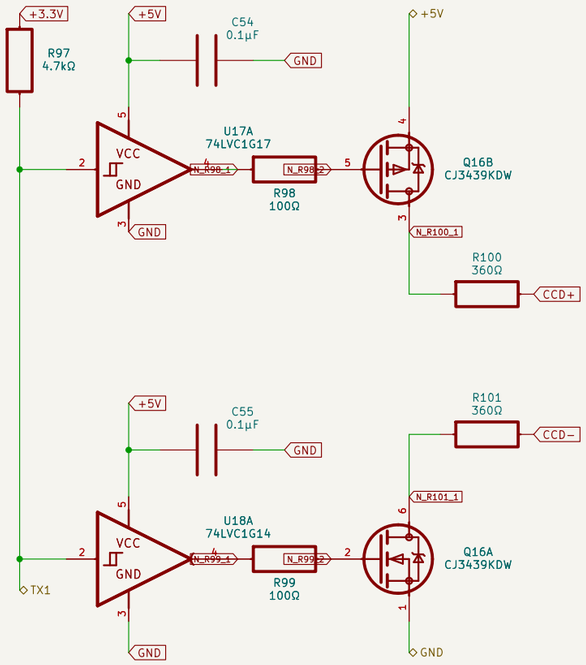
Matching signal names connect between sheets. Values and package pins follow the atopile netlist.

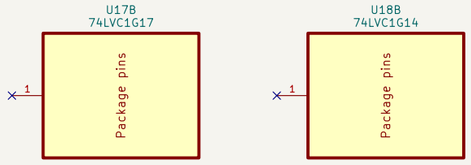










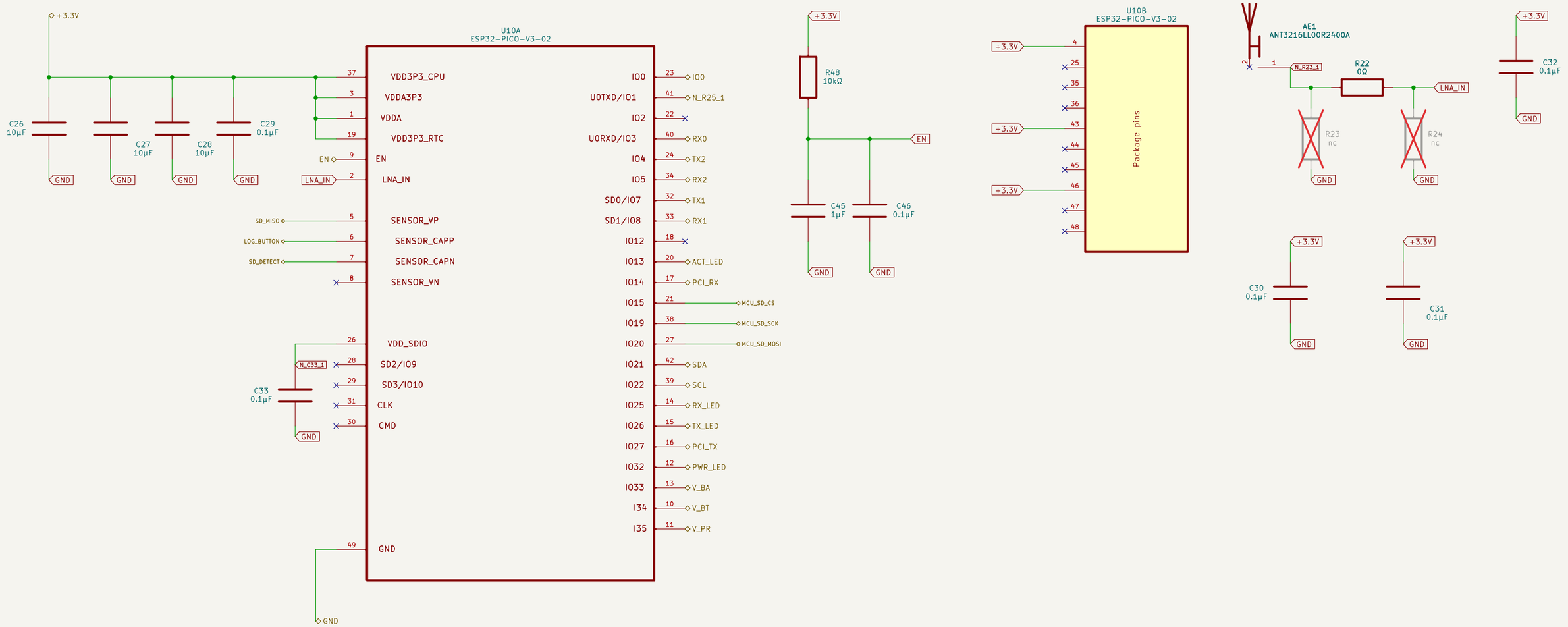


Atopile source / editable KiCad view / passive pins require electrical review.

ChryslerScanner / atopile		
Sheet: /CCD interface/		
File: 01_ccd.kicad_sch		
Title: CCD interface		
Size: A3	Date: 2026-10-13	**Rev: V240-C/OBD**
KiCad E.D.A. 10.0.5		Id: 3/22

ESP32 + RF

Matching signal names connect between sheets. Values and package pins follow the atopile netlist.



SD revision: GPIO19/20 SPI, GPIO36 MISO, GPIO15 CS, GPIO37 button, GPIO38 detect. Firmware deferred.

Assembly option: R23 and R24 are DNP.

Atopile source / editable KiCad view / passive pins require electrical review.

ChryslerScanner / atopile

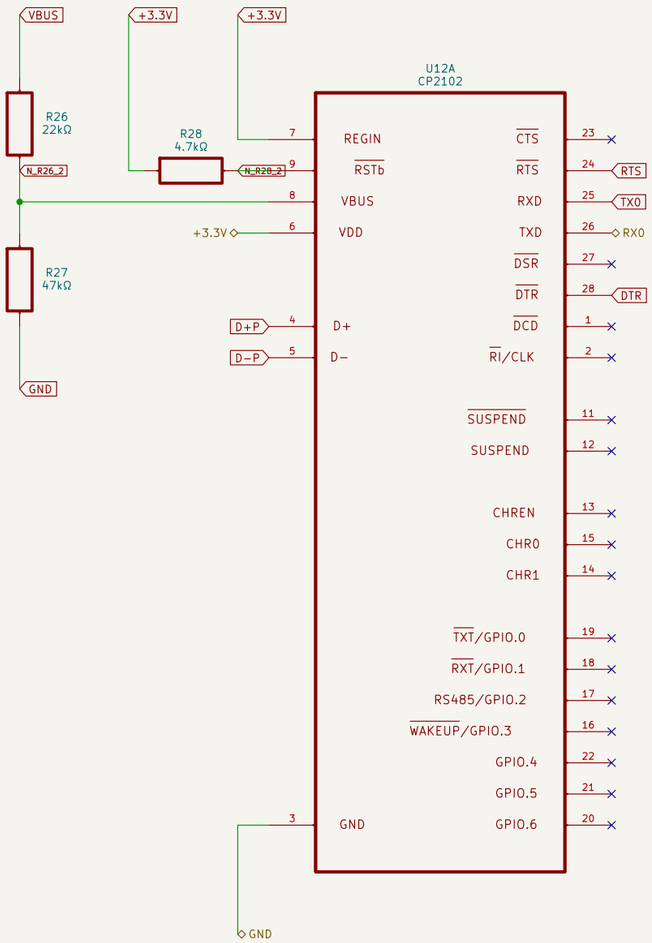
Sheet: /ESP32 + RF/
File: 16_mcu.kicad_sch

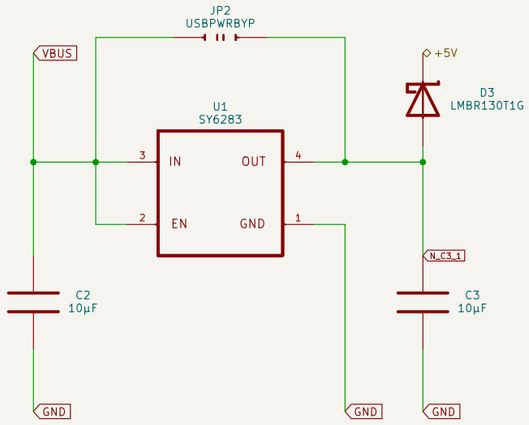
Title: ESP32 + RF

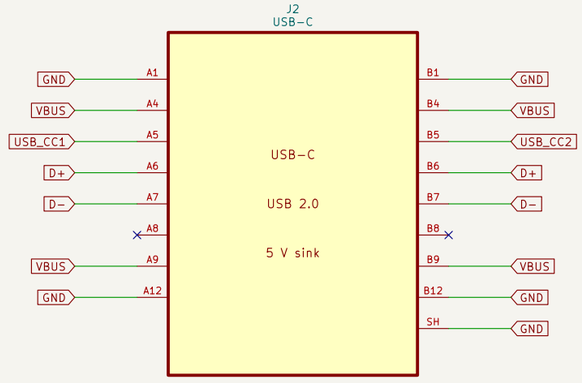
Size: A3	Date: 2026-10-13	**Rev: V240-C/OBD**
KiCad E.D.A. 10.0.5		Id: 4/22

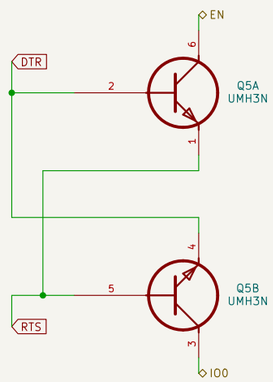
USB + reset / boot

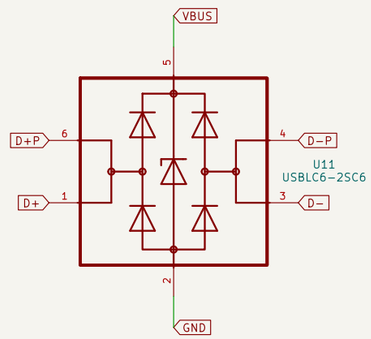
Matching signal names connect between sheets. Values and package pins follow the atopile netlist.

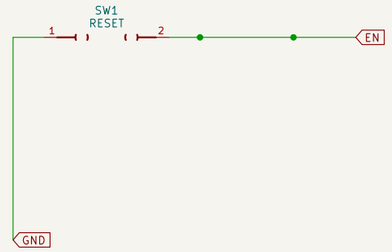


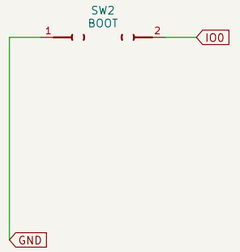


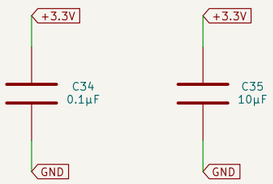


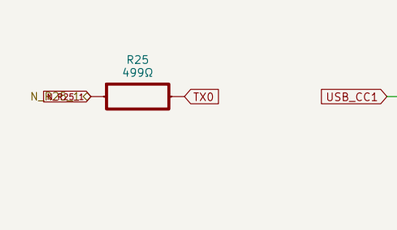


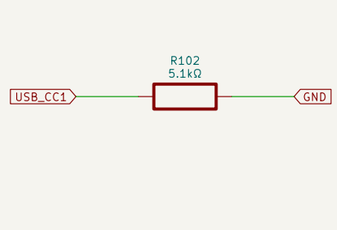


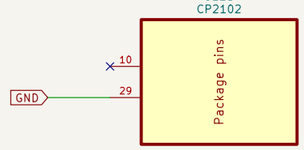












USB−C: USB 2.0 device / 5 V sink. R102 and R103: independent 5.1 kohm CC pull−downs.

Atopile source / editable KiCad view / passive pins require electrical review.

ChryslerScanner / atopile

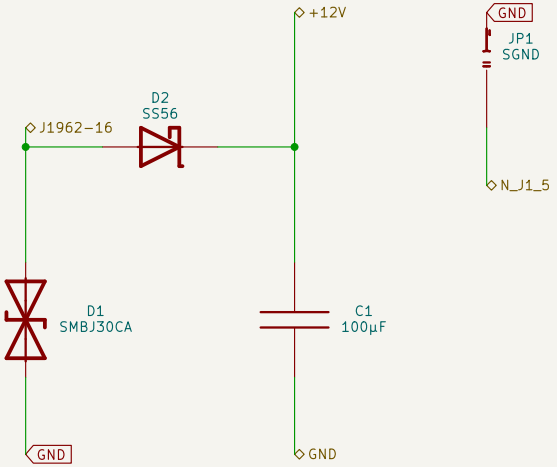
Sheet: /USB + reset / boot/
File: 19_usb.kicad_sch

Title: USB + reset / boot

Size: A3	Date: 2026−10−13	**Rev: V240−C/OBD**
KiCad E.D.A. 10.0.5		Id: 5/22

Vehicle power input

Matching signal names connect between sheets. Values and package pins follow the atopile netlist.



Atopile source / editable KiCad view / passive pins require electrical review.

ChryslerScanner / atopile

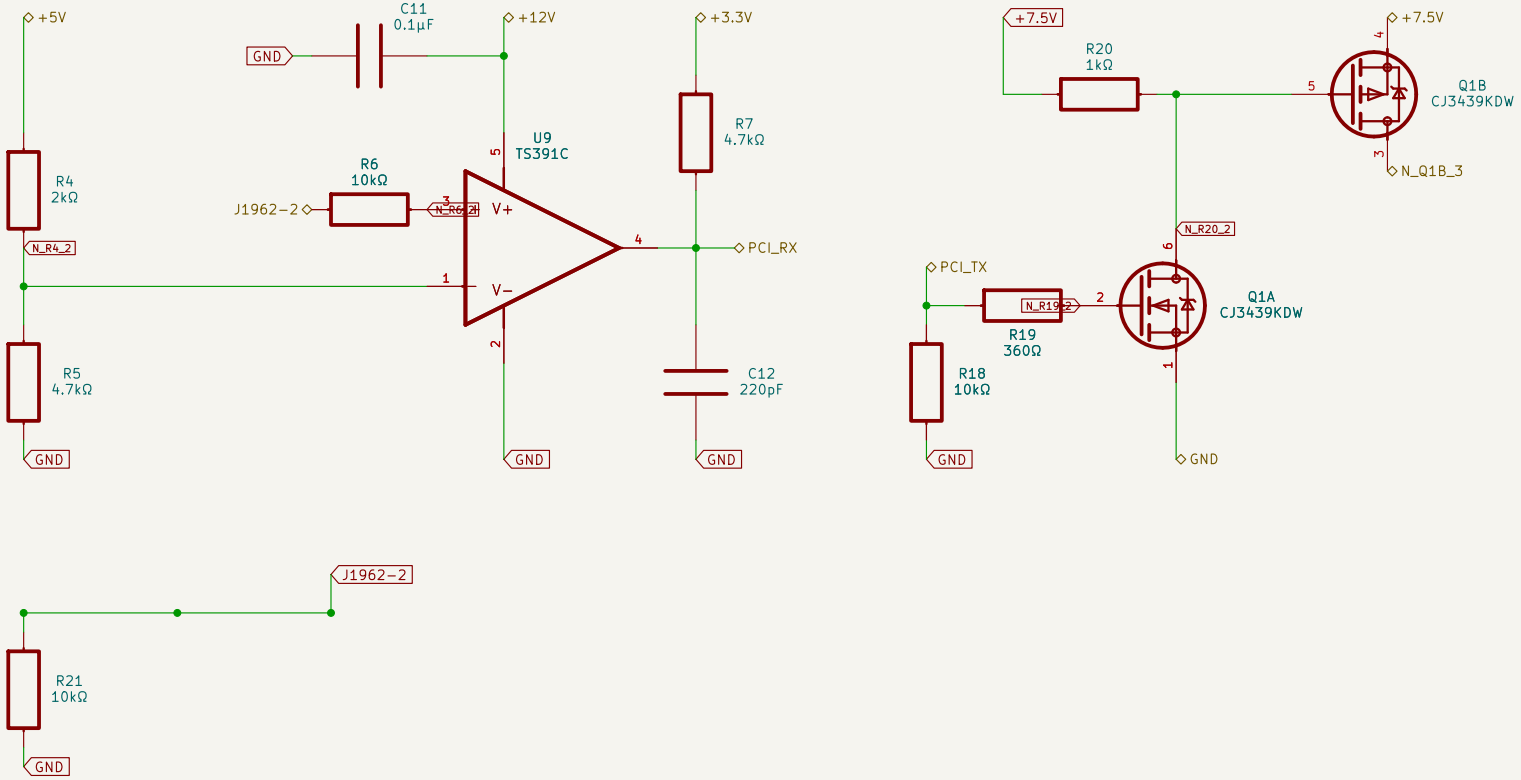
Sheet: /Vehicle power input/
File: 07_vehicle-input.kicad_sch

Title: Vehicle power input

Size: A4	Date: 2026-10-13	**Rev: V240-C/OBD**
KiCad E.D.A. 10.0.5		Id: 6/22

PCI / J1850

Matching signal names connect between sheets. Values and package pins follow the atopile netlist.

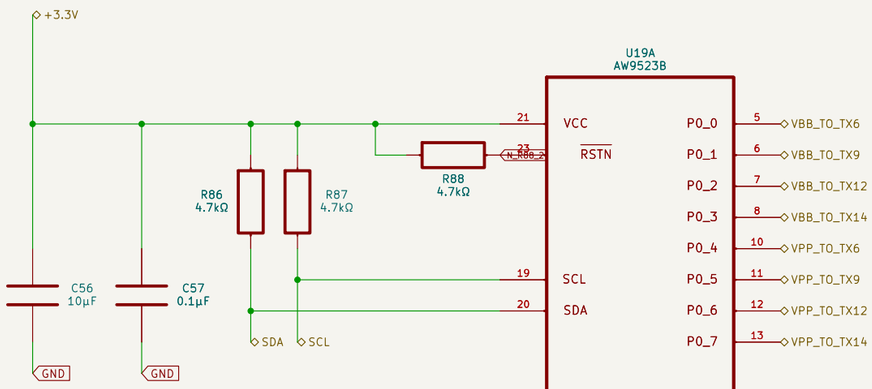


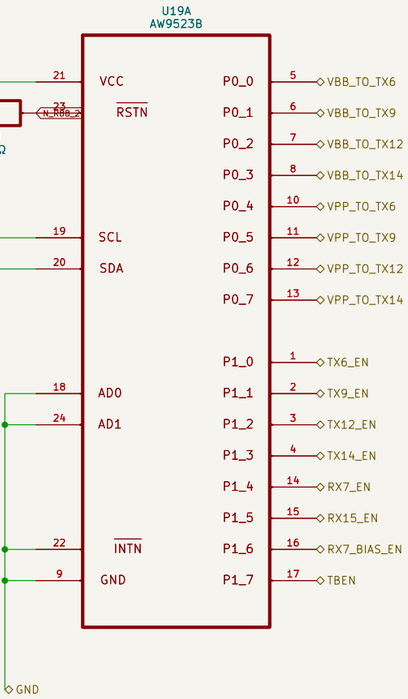
Atopile source / editable KiCad view / passive pins require electrical review.

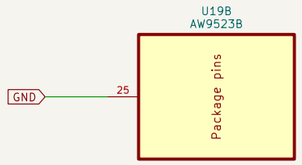
ChryslerScanner / atopile		
Sheet: /PCI / J1850/		
File: 02_pci.kicad_sch		
Title: PCI / J1850		
Size: A4	Date: 2026-10-13	**Rev: V240-C/OBD**
KiCad E.D.A. 10.0.5		Id: 7/22

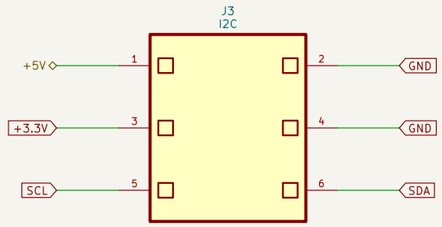
I/O expander + I2C

Matching signal names connect between sheets. Values and package pins follow the atopile netlist.









Atopile source / editable KiCad view / passive pins require electrical review.

ChryslerScanner / atopile

Sheet: /I/O expander + I2C/
File: 18_control.kicad_sch

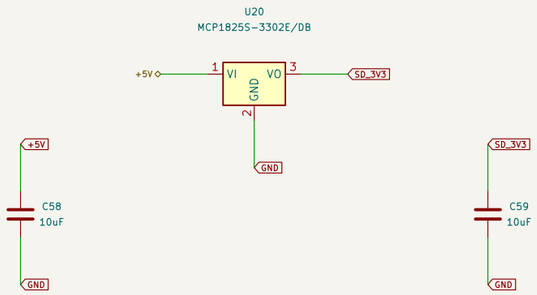
Title: I/O expander + I2C

Size: A3	Date: 2026-10-13	**Rev: V240-C/OBD**
KiCad E.D.A. 10.0.5		Id: 8/22

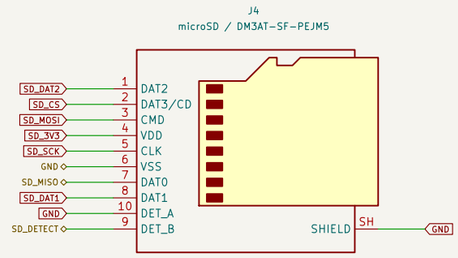
On-board microSD logger (SPI)

Existing ESP32-PICO-V3-02 | removable storage | native scanner functions retained | firmware deferred

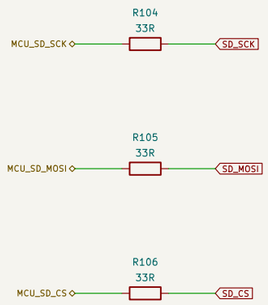
Dedicated 3.3 V SD supply from the existing +5 V rail



J4: Hirose DM3AT-SF-PEJM5 / SPI mode



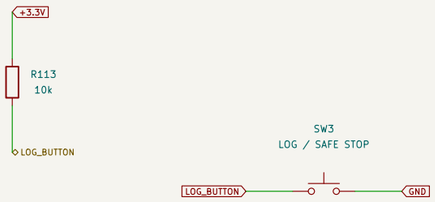
MCU-side series damping



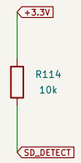
SD CMD/DAT pull-ups / MCU CS strap bias



Active-low record / safe-stop input



Card-detect input



GPIO15 CS is biased high at reset; GPIO12 and GPIO2 remain unused.

GPIO36/37/38 are input-only and have external pull-ups.

GPIO20 is package-specific; use ESP-IDF support for ESP32-PICO-V3.

SD rail transient current and total +5 V budget need hardware validation.

Bulk capacitance is not a safe-power-off hold-up supply.

Firmware deferred: this hardware revision does not yet record data.

Future status indication reuses existing ACT_LED / GPIO13.

Existing sheets use passive visualization pins; native ERC is not independent certification of the scanner circuit.

ChryslerScanner / local hardware revision

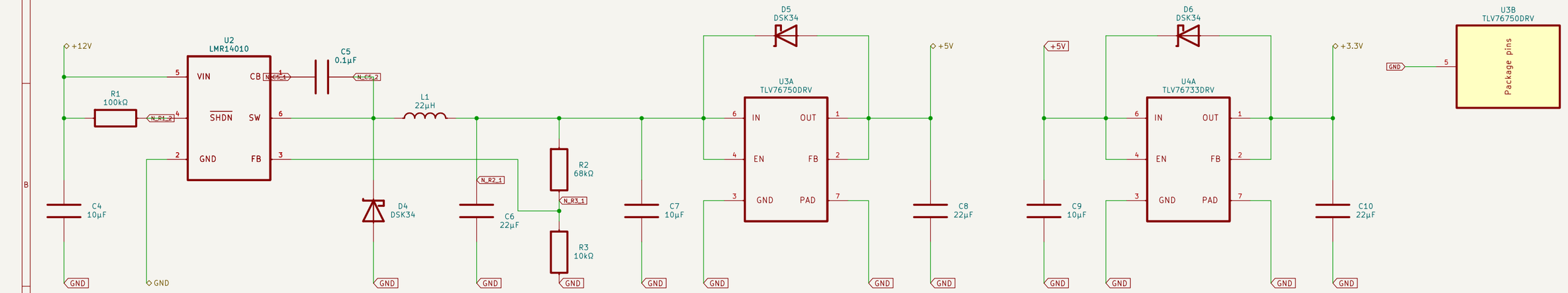
Sheet: /Integrated SD logger/
File: 22_sd-logger.kicad_sch

Title: Integrated microSD logging hardware

Size: A3	Date: 2026-10-20	**Rev: V240-SD-A**
KiCad E.D.A. 10.0.5		Id: 9/22

5 V / 3.3 V supplies

Matching signal names connect between sheets. Values and package pins follow the atopile netlist.



REVIEW: U3 / U4 pin 5 is NC in the source; TLV767 DRV requires GND.

Source connectivity is preserved here. Correct this in atopile before routing.

Atopile source / editable KiCad view / passive pins require electrical review.

ChryslerScanner / atopile

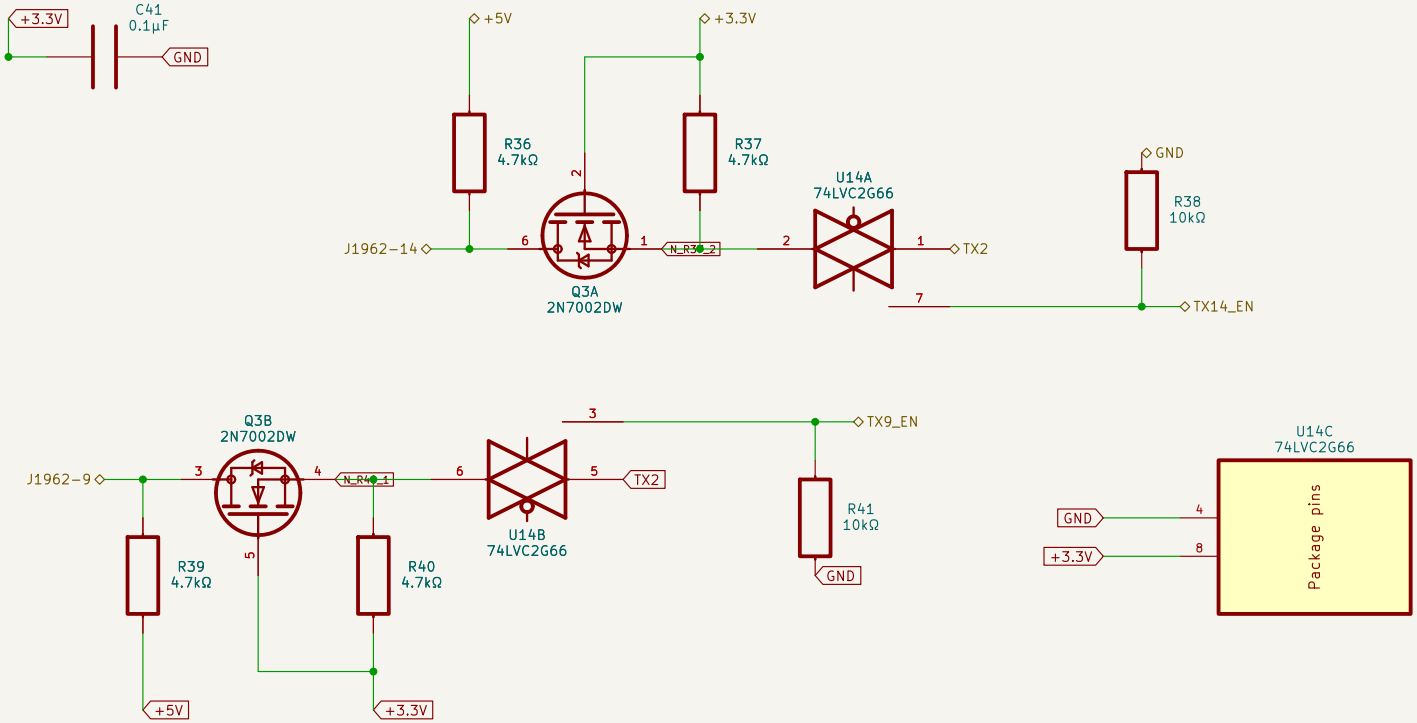
Sheet: /5 V / 3.3 V supplies/
File: 08_power-5v.kicad_sch

Title: 5 V / 3.3 V supplies

Size: A3	Date: 2026-10-13	**Rev: V240-C/OBD**
KiCad E.D.A. 10.0.5		Id: 10/22

SCI TX14 / TX9

Matching signal names connect between sheets. Values and package pins follow the atopile netlist.

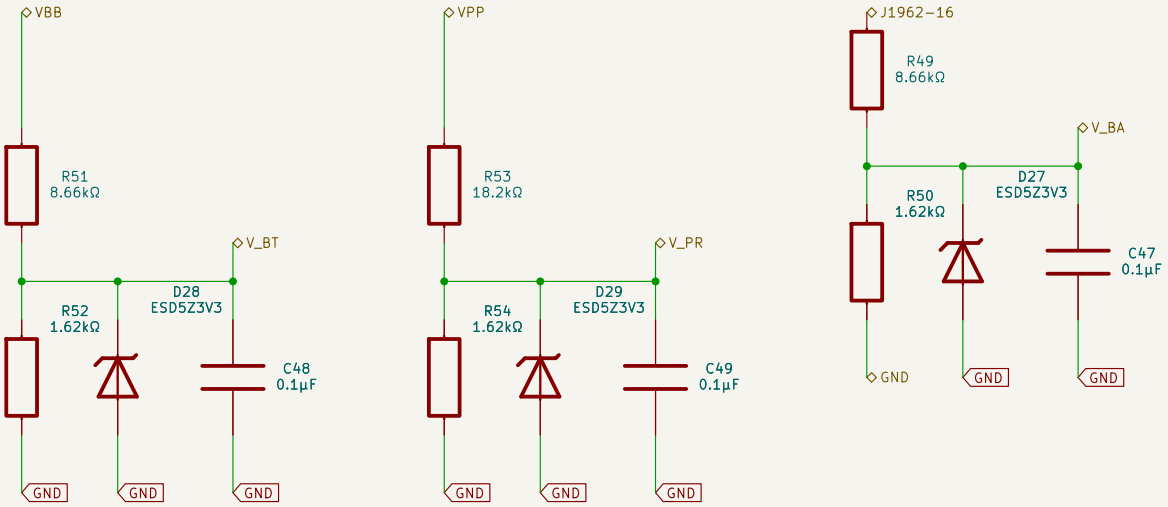


Atopile source / editable KiCad view / passive pins require electrical review.

ChryslerScanner / atopile		
Sheet: /SCI TX14 / TX9/		
File: 03_sci-14-9.kicad_sch		
Title: SCI TX14 / TX9		
Size: A4	Date: 2026-10-13	**Rev: V240-C/OBD**
KiCad E.D.A. 10.0.5		Id: 11/22

Voltage monitoring

Matching signal names connect between sheets. Values and package pins follow the atopile netlist.

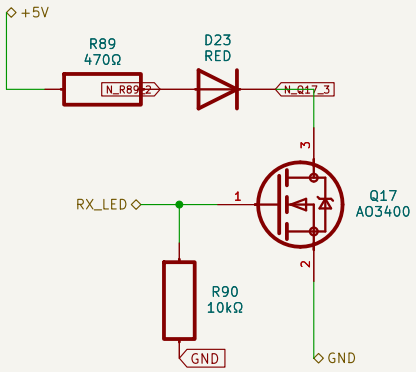


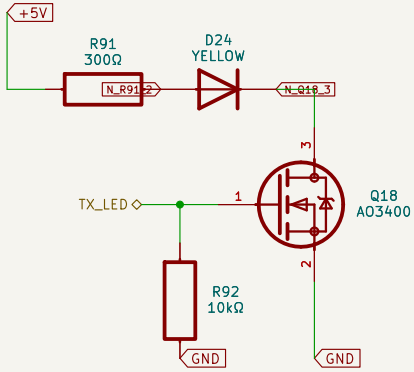
Atopile source / editable KiCad view / passive pins require electrical review.

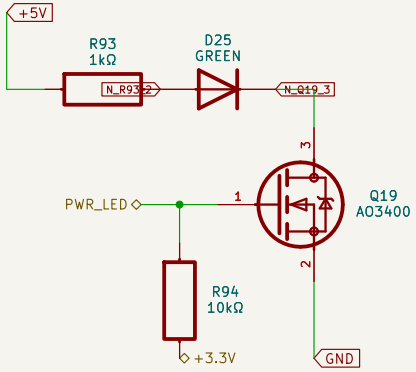
ChryslerScanner / atopile		
Sheet: /Voltage monitoring/ File: 17_adc.kicad_sch		
Title: Voltage monitoring		
Size: A4	Date: 2026-10-13	**Rev: V240-C/OBD**
KiCad E.D.A. 10.0.5		Id: 12/22

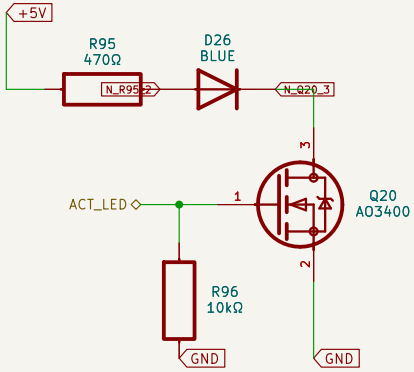
Status LED drivers

Matching signal names connect between sheets. Values and package pins follow the atopile netlist.









Atopile source / editable KiCad view / passive pins require electrical review.

ChryslerScanner / atopile

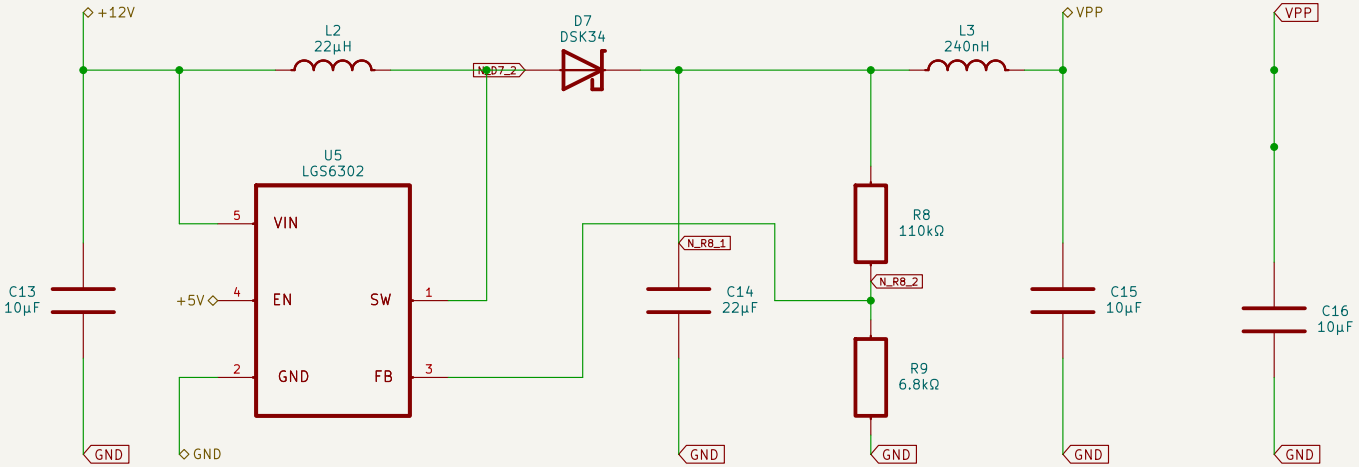
Sheet: /Status LED drivers/
File: 20_leds.kicad_sch

Title: Status LED drivers

Size: A4	Date: 2026-10-13	**Rev: V240-C/OBD**
KiCad E.D.A. 10.0.5		Id: 13/22

VPP boost supply

Matching signal names connect between sheets. Values and package pins follow the atopile netlist.

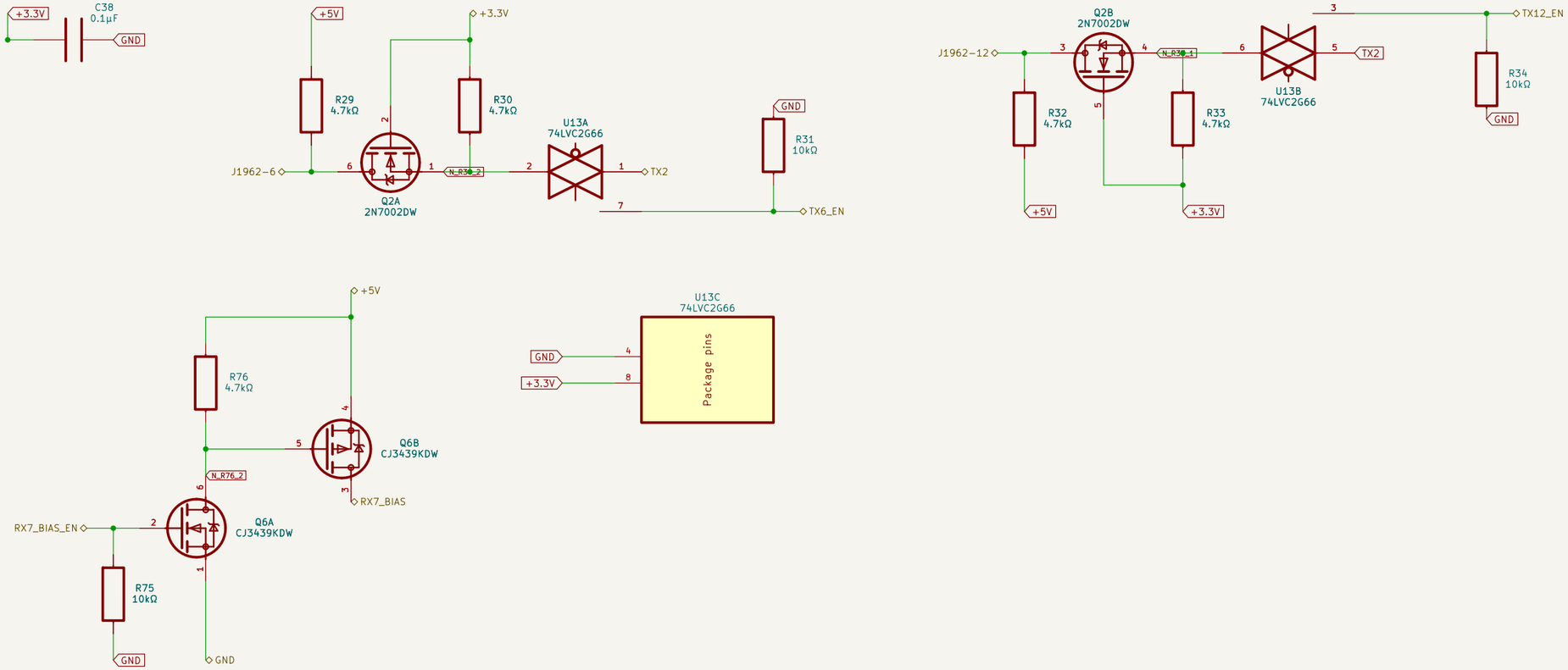


Atopile source / editable KiCad view / passive pins require electrical review.

ChryslerScanner / atopile		
Sheet: /VPP boost supply/ File: 09_power-vpp.kicad_sch		
Title: VPP boost supply		
Size: A4	Date: 2026-10-13	**Rev: V240-C/OBD**
KiCad E.D.A. 10.0.5		Id: 14/22

SCI TX6 / TX12 + RX bias

Matching signal names connect between sheets. Values and package pins follow the atopile netlist.



Atopile source / editable KiCad view / passive pins require electrical review.

ChryslerScanner / atopile

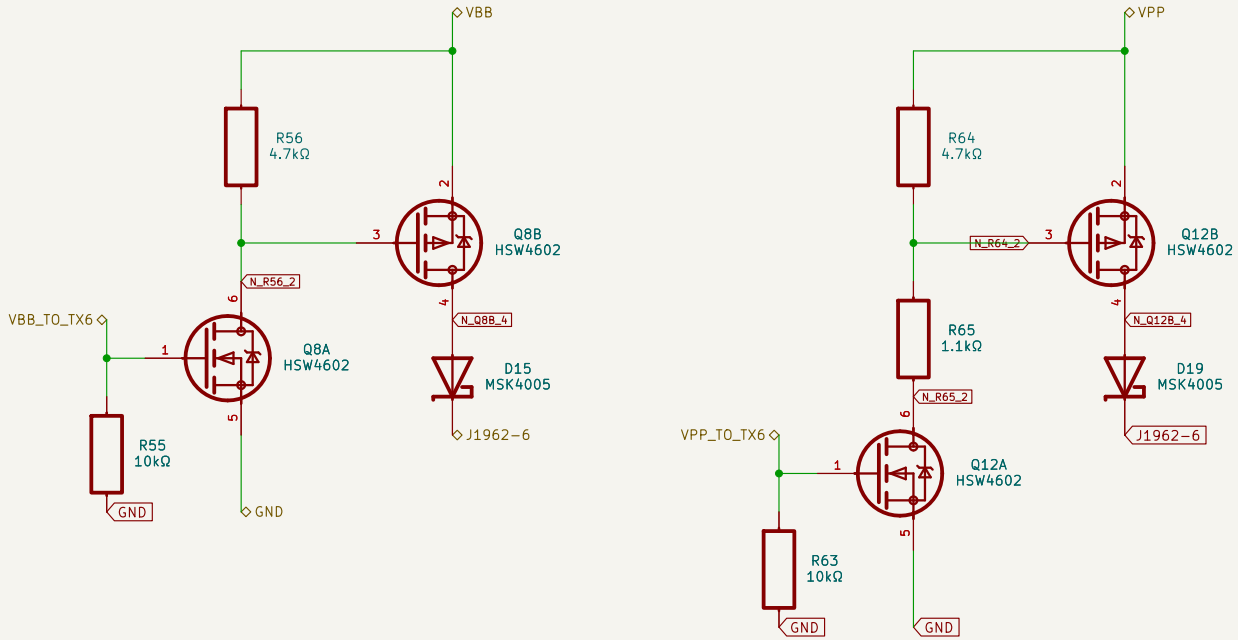
Sheet: /SCI TX6 / TX12 + RX bias/
File: 04_sci-6-12.kicad_sch

Title: SCI TX6 / TX12 + RX bias

Size: A3	Date: 2026-10-13	**Rev: V240-C/OBD**
KiCad E.D.A. 10.0.5		Id: 15/22

SCI TX6 VBB / VPP drivers

Matching signal names connect between sheets. Values and package pins follow the atopile netlist.



Atopile source / editable KiCad view / passive pins require electrical review.

ChryslerScanner / atopile

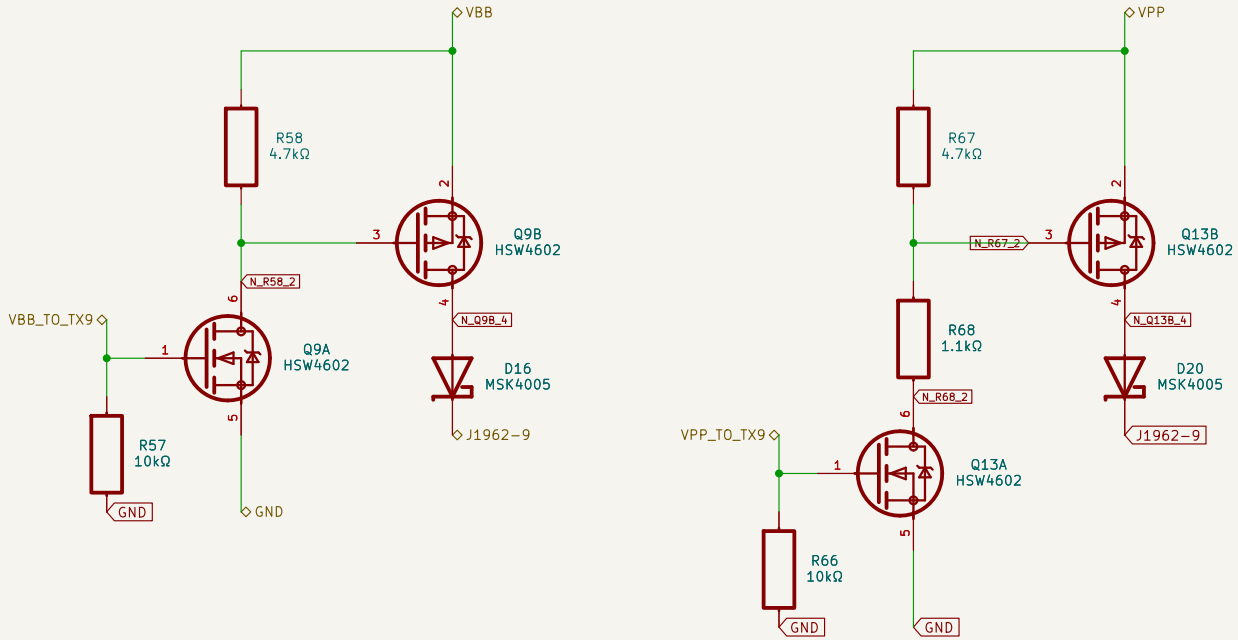
Sheet: /SCI TX6 VBB / VPP drivers/
File: 12_program-6.kicad_sch

Title: SCI TX6 VBB / VPP drivers

Size: A4	Date: 2026-10-13	**Rev: V240-C/OBD**
KiCad E.D.A. 10.0.5		Id: 16/22

SCI TX9 VBB / VPP drivers

Matching signal names connect between sheets. Values and package pins follow the atopile netlist.



Atopile source / editable KiCad view / passive pins require electrical review.

ChryslerScanner / atopile

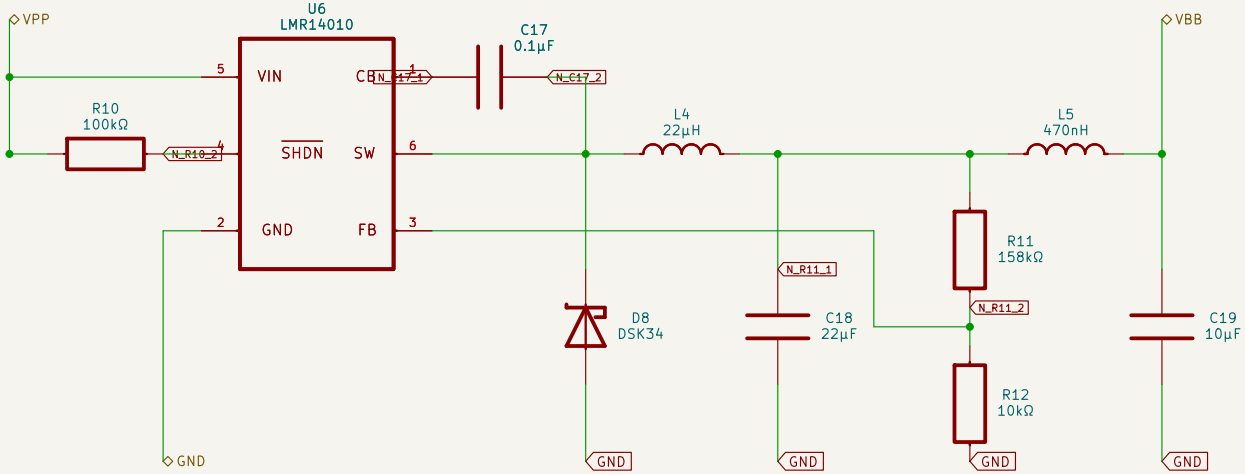
Sheet: /SCI TX9 VBB / VPP drivers/
File: 13_program-9.kicad_sch

Title: SCI TX9 VBB / VPP drivers

Size: A4	Date: 2026-10-13	**Rev: V240-C/OBD**
KiCad E.D.A. 10.0.5		Id: 17/22

VBB supply

Matching signal names connect between sheets. Values and package pins follow the atopile netlist.

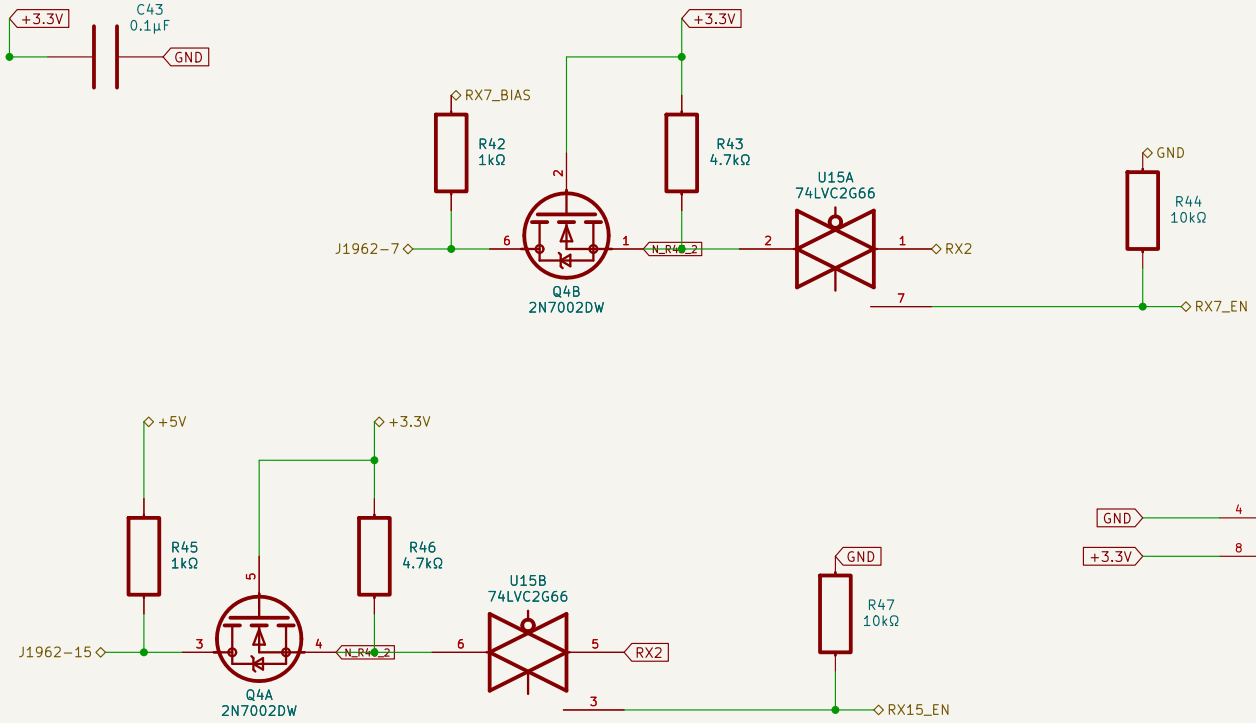


Atopile source / editable KiCad view / passive pins require electrical review.

ChryslerScanner / atopile		
Sheet: /VBB supply/ File: 10_power-vbb.kicad_sch		
Title: VBB supply		
Size: A4	Date: 2026-10-13	**Rev: V240-C/OBD**
KiCad E.D.A. 10.0.5		Id: 18/22

SCI RX7 / RX15

Matching signal names connect between sheets. Values and package pins follow the atopile netlist.

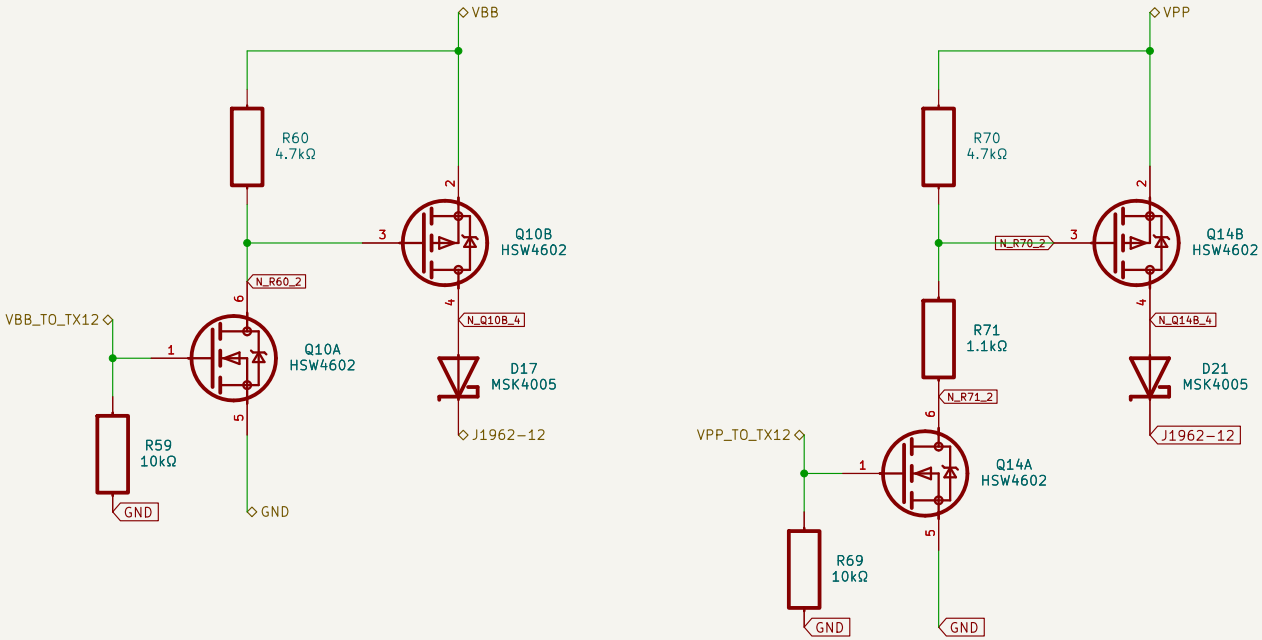


Atopile source / editable KiCad view / passive pins require electrical review.

ChryslerScanner / atopile		
Sheet: /SCI RX7 / RX15/		
File: 05_sci-rx.kicad_sch		
Title: SCI RX7 / RX15		
Size: A4	Date: 2026-10-13	**Rev: V240-C/OBD**
KiCad E.D.A. 10.0.5		Id: 19/22

SCI TX12 VBB / VPP drivers

Matching signal names connect between sheets. Values and package pins follow the atopile netlist.



Atopile source / editable KiCad view / passive pins require electrical review.

ChryslerScanner / atopile

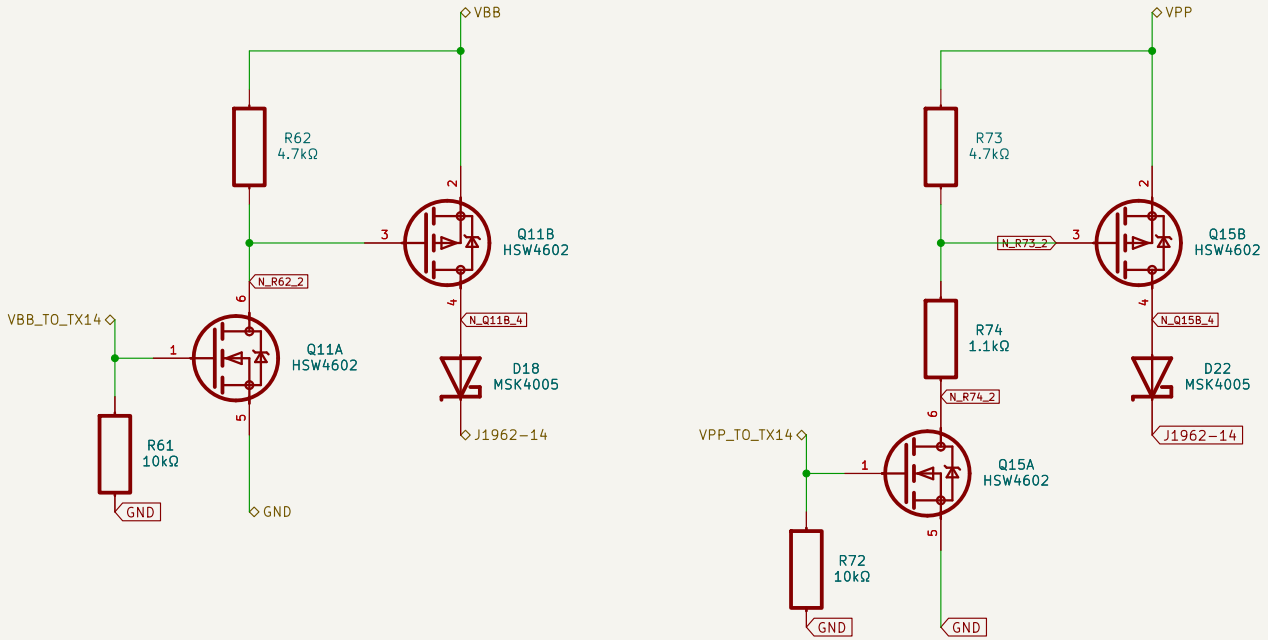
Sheet: /SCI TX12 VBB / VPP drivers/
File: 14_program-12.kicad_sch

Title: SCI TX12 VBB / VPP drivers

Size: A4	Date: 2026-10-13	**Rev: V240-C/OBD**
KiCad E.D.A. 10.0.5		Id: 20/22

SCI TX14 VBB / VPP drivers

Matching signal names connect between sheets. Values and package pins follow the atopile netlist.



Atopile source / editable KiCad view / passive pins require electrical review.

ChryslerScanner / atopile

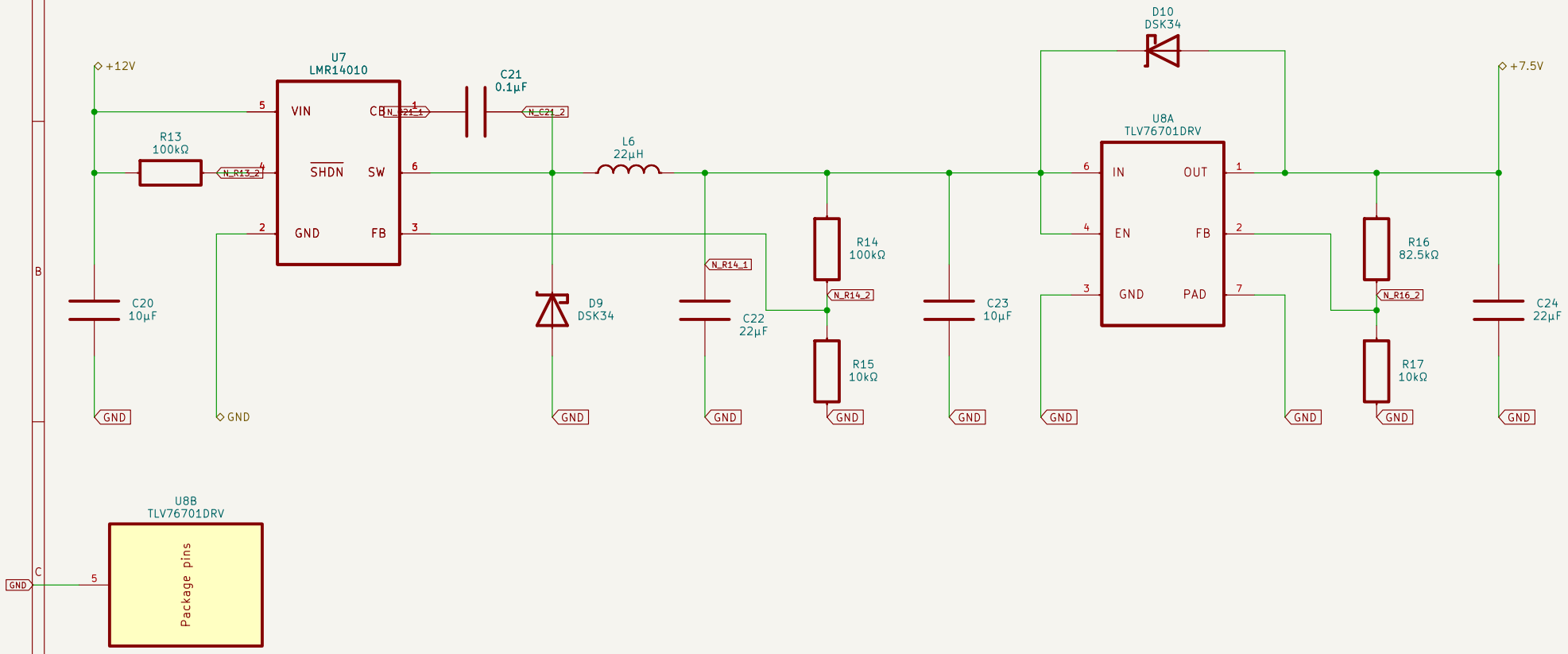
Sheet: /SCI TX14 VBB / VPP drivers/
File: 15_program-14.kicad_sch

Title: SCI TX14 VBB / VPP drivers

Size: A4	Date: 2026-10-13	**Rev: V240-C/OBD**
KiCad E.D.A. 10.0.5		Id: 21/22

7.5 V supply

Matching signal names connect between sheets. Values and package pins follow the atopile netlist.



REVIEW: U8 pin 5 is NC in the source; TLV767 DRV requires GND.

Source connectivity is preserved here. Correct this in atopile before routing.

Atopile source / editable KiCad view / passive pins require electrical review.

ChryslerScanner / atopile

Sheet: /7.5 V supply/
File: 11_power-7v5.kicad_sch

Title: 7.5 V supply

Size: A4	Date: 2026-10-13	**Rev: V240-C/OBD**
KiCad E.D.A. 10.0.5		Id: 22/22